Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Cubes and Cuboids: Faces, Edges and Vertices

A **three dimensional shape** has length, breadth and height, so it takes up space. That is why these shapes are also called **solid shapes**. The cube and the cuboid are two of these solid shapes. Before we count their parts, let us fix three words you will need.

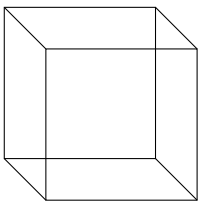
- A **face** (or surface) is one of the sides of the shape. A face may be flat or curved.
- An **edge** is a line where two faces meet. It may be straight or curved.
- A **vertex** is a point corner where edges meet. The plural of vertex is **vertices**.

The cube. A cube is a solid shape whose faces are all equal squares. Look at **fig-01**: a cube drawn as a see-through solid, with one square offset from another and their corners joined. Count its parts carefully.

- It has **6 square faces** — the six equal squares that cover it.
- It has **12 edges** — the straight lines where the faces meet.
- It has **8 vertices** — the 8 point corners where the edges meet.

A cube is a **regular** solid shape: its faces are all the same shape and size.

Visual



A cube drawn as a see-through solid: two equal squares, the nearer one offset down and to the right of the farther one, with the four pairs of corresponding corners joined.

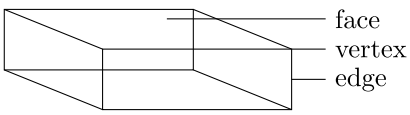
The cuboid. A cuboid is also a solid shape with six faces, but its faces are rectangles rather than equal squares. In **fig-02** the cuboid is drawn wide and shallow, and three leader lines name its parts: the top **face**, a **vertex** (the point corner), and an **edge** (the line where two faces meet).

Count its parts in the same way:

- It has **6 faces**.
- It has **12 edges**.
- It has **8 vertices**.

The cube and the cuboid share the same numbers of faces, edges and vertices. What makes them different is that every face of a cube is an equal square, while a cuboid's faces are rectangles (usually of two or three different sizes).

Visual



A cuboid drawn the same way, wide and shallow, with three leader lines to the right naming the top face, the nearer top-right corner and the nearer right-hand vertical edge.

Practice

1. How many vertices has a cuboid?
2. How many edges has a cube?

3. How many square faces has a cube, and how are a cube's faces different from a cuboid's faces?

Section 2: Pyramids and Cones: Faces, Edges and Vertices

A pyramid is named by the shape of its base. The base is the flat face it stands on, and from every corner of that base one straight edge runs up to a single point at the top, called the apex. Look at the triangular base pyramid in fig-03: its base is a triangle, so it is a triangular based pyramid. Its special name is a tetrahedron.

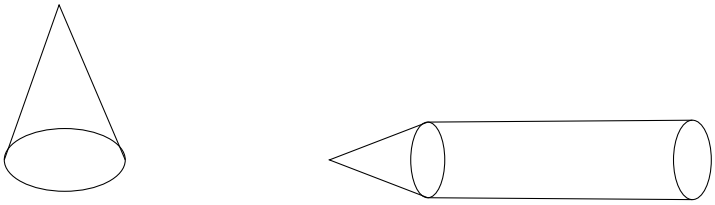
The square based pyramid in fig-04, the pentagonal based pyramid in fig-05 and the rectangular based pyramid in fig-06 are all named the same way — by what their base is.

Every pyramid has faces, edges and vertices, and you can count them. Remember: a face is a flat surface, an edge is where two faces meet, and a vertex is a corner where edges meet. Here is how the count works out:

Shape	Faces	Edges	Vertices
Triangular based pyramid	4	6	4
Square based pyramid	5	8	5
Pentagonal based pyramid	6	10	6
Rectangular based pyramid	5	8	5

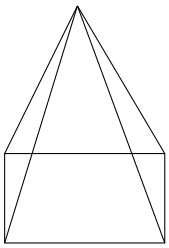
A cone (fig-07) is different. It is round, so it has a flat circular base and a curved face that narrows up to one point. Altogether a cone has two faces — one curved face and one flat face — one edge (the rim of the circular base), and one vertex (the point at the top).

Visual



Two drawings side by side. On the left an upright cone: an elliptical base with the two slant sides running up from the ends of the ellipse to the apex. On the right a cone lying on its side, apex at the left, its two slant sides reaching a narrow upright ellipse whose top and bottom then continue as two parallel lines to a second narrow ellipse further right.

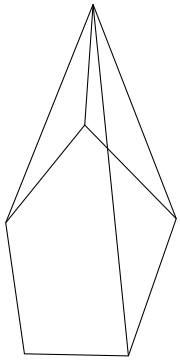
Visual



Rectangular based Pyramid

A rectangular based pyramid, drawn like the square one but on a base wider than it is tall.

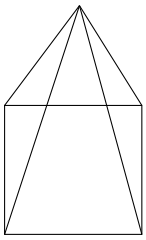
Visual



Pentagonal based Pyramid

A pentagonal based pyramid. The base is drawn as a pentagon standing on one side with its fifth vertex uppermost, and the apex above is joined to that vertex, to the two vertices either side of it, and to the lower-right vertex — four edges from the apex, the last of them crossing the face of the solid.

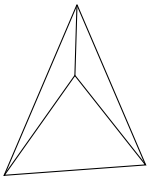
Visual



Square base Pyramid

A square base pyramid. The base is drawn as an upright rectangle and the apex above is joined to all four of its corners, so two edges run outside the rectangle and two cross it.

Visual



Triangular base Pyramid (Tetrahedron)

A triangular base pyramid. The base triangle is drawn in perspective, its third vertex raised into the middle of the figure, and the apex above is joined to all three base vertices.

Let's name a solid and then count its parts.

Example 1 — Name the solid with a square base. Its base is a square, so it is a square based pyramid. From the table, a square based pyramid has 5 faces, 8 edges and 5 vertices.

Example 2 — How many edges does a pentagonal based pyramid have? Its base is a pentagon (five sides). Reading the row for the pentagonal based pyramid: 6 faces, 10 edges, 6 vertices. So it has 10 edges.

Example 3 — A cone has a curved face and a flat circular base. The flat base meets the curved face along one circular rim. So a cone has 2 faces, 1 edge and 1 vertex.

Practice

1. Draw a hexagonal base pyramid (a pyramid whose base has six sides) and list its properties. How many faces, edges and vertices does it have?
2. How many vertices does a tetrahedron have? What kind of base does it have, and how is it named?

3. How many faces, edges and vertices does a cone have?

Check your work

Attempt every question before you read this page.

Cubes and Cuboids: Faces, Edges and Vertices

1. A cuboid has 8 vertices.
2. A cube has 12 edges.
3. A cube has 6 square faces. A cube's faces are all equal squares, while a cuboid's faces are rectangles (so they are not all equal squares).

Pyramids and Cones: Faces, Edges and Vertices

1. A hexagonal based pyramid has 7 faces, 12 edges and 7 vertices.

   The pyramid is named by its base, so a hexagonal base gives a hexagonal based pyramid. Its base is a hexagon (6 sides). A pyramid with a base of n sides has $n + 1$ faces (the n triangular side faces plus the base), $2n$ edges, and $n + 1$ vertices. With $n = 6$: $6 + 1 = 7$ faces, $2 \times 6 = 12$ edges, and $6 + 1 = 7$ vertices.
2. A tetrahedron has 4 vertices. It has a triangular base, so it is also called a triangular based pyramid.

   A tetrahedron is a triangular based pyramid, and the triangular based pyramid row of the table shows 4 faces, 6 edges and 4 vertices — so it has 4 vertices.
3. A cone has 2 faces (one curved face and one flat circular base), 1 edge and 1 vertex.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-a0bda39a	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____